Pavanitha Manche

Software Engineer
Computer Science and Engineering

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Education

Indian Institute of Technology(IIT), Dharwad 2020-2024
Btech in Computer Science Engineering (CPI - 8.74) *Dharwad, Karnataka*

- **Relevant Coursework:** Data Structures and Algorithms, Software System Labs, Design and Analysis of Algorithms, Artificial Intelligence, Discrete Structures, Computer Architecture, Databases and Information Systems, Software Development for scientific Computing, Operating Systems, Computer Networks, Compilers

Experience

ABB Innovation Centre May 2024 – present
Software Engineer | C#, .NET, Angular.js, Kubernetes *Bangalore, Karnataka*

- Designed an automated digital solution for the gas industry, aimed at optimizing gas-lift setpoints.
- Developed backend services integrating a mathematical model with ABB's platform services using C# and .NET .
- Supported the application deployment in Kubernetes clusters for scalability and resilience.
- Developed a reusable MQTT adapter to transform client telemetry data, enabling integration with any digital solutions across organization.

Dvara Solutions Aug. 2023 – Apr. 2024
AI/ML intern | Python, react.js, Langchain, postgres, Docker, jenkins *Dharwad, Karnataka*

- Developed a voice-based AI chatbot assistant using RAG framework with a focus on rural financial services specifically for users communicating in their native lnaguages, leveraging generative AI and Large Language Models (LLMs) .
- Developed a backend framework integrating LLMs with vector database ChromaDB for efficient similarity-based queries.
- Enhanced chatbot performance through prompt engineering and implemented session memory to maintain conversation context.
- Designed a comprehensive framework that integrates ASR and translation techniques for native voice input conversion before sending it to an LLM, and used TTS to convert the response back into native language audio, employing models like Whisper and Google Translate.

IIT Dharwad Jan. 2023 - Apr. 2024
Teaching Assistant *Dharwad, Karnataka*

- Worked as a teaching assistant for the following courses: Introduction to C, Introduction to Python, Software Systems Lab, and Data Structures and Algorithms.

Projects

Voice OTP based Authentication System | *Machine Learning*

- Developed a two-factor Voice OTP authentication system incorporating speaker identification and digit recognition.
- Designed a VOA framework using ECAPA-TDNN for speaker representation and a conformer-based model for digit recognition, achieving 93.21% accuracy through in-domain fine-tuning.
- Evaluated deployment feasibility by measuring inference times and memory requirements across three devices, optimizing suitability for resource-constrained environments like mobile phones.

Building Custom Search Engine | *Python, react, flask, ElasticSearch*

- Developed a specialized search engine for government websites based on a problem statement from Niti Aayog, addressing gaps overlooked by mainstream search engines.
- Employed web scraping, indexing, crawling, and Natural Language Processing (NLP) techniques, leveraging Elasticsearch efficient data storage and visualization of indexed information.

Publications

- **Design and Development of Voice OTP Authentication System**
  Pavanitha Manche, Sahaja Nandyala, Jagabandhu Mishra, Gayathri Ananthanarayanan, S. R. Mahadeva Prasanna
  25th International Conference on Speech and Computer SPECOM 2023 |
- **Speaker and Digit Representation based Voice OTP System**
  Sahaja Nandyala, **Pavanitha Manche**, Jagabandhu Mishra, S. R. Mahadeva Prasanna
  2024 International Conference on Signal Processing and Communications (SPCOM) |

Technologies

- **Programming:** Python, C, C++, C#, Java
- **Frontend :** React.js, angular.js, HTML, CSS, javascript
- **Backend :** Node.js, PHP, MongoDB, MySQL, postgres
- **Tools/frameworks :** Tensorflow, Keras, Git, docker, jenkins, kubernetes
- **Operating Systems** Windows, Linux

Certifications

- Convolutional Neural Networks Certificate
- CNN in tensorflow certificate
- Natural Language Processing in TensorFlow certificate

Achievements

- Secured 3rd position in both AI Trek and Algostrike 3.0 competitions during the Parsec Technical Fest 2023 at IIT Dharwad.
- Advanced to the final round in Convolve, a Pan-IIT AI & ML Hackathon.
- Secured a position among the top 30 in the CodeRush Challenge conducted by AlgoUniversity at IIT Dharwad.
- Actively contributed to the E-Summit 2022 at IIT Dharwad as a core Event Management team member under the Institute Innovation Council.
- Engaged in the Fine Arts Club, Coding Club and Photography and Films club at IIT Dharwad, fostering creative initiatives within the institute.